Ram Calculus and Circle-Domain Uniformization

M149: Sharper Stream Bounds, Controlled Metric Distortion, and Period-Preserving Certificates for Koebe's Problem

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Abstract

The September 23 manuscript *Koebe's Circle-Domain Conjecture*, released in OpenAI's October 6 mathematics collection, claims unrestricted circle-domain uniformization. Its finite-transfer and stream-coordinate steps have direct relevance to the physical moment, compatible differential-form, and full-component certificate methods in the Ram Calculus corpus. We rederive these interfaces and prove a sharper finite-completion Stokes estimate: the nonsingular-hole coefficient 2π can be replaced by 2 using exact variation of distance on a circle. A joint area–radius budget further replaces separately maximized energy and boundary contributions. We derive the exact pointwise minimum of coordinate energy under a prescribed metric condition-number cap, quantify its perturbation stability, and give the minimum-physical-energy correction for a finite complete period bank. We also state and prove a robust full-component compactness theorem and explain the remaining all-scale requirements. These are useful local and conditional results. They neither independently establish the source's unrestricted theorem nor certify its measurable path-law fusion and transfinite selection arguments. The report supplies explicit proofs, source-step comparisons, reproducible finite checks, and a research program that keeps those distinctions visible.

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1 Problem, Sources, and Exact Scope

1.1 The target and its quantifiers

A domain is a nonempty connected open subset of $\widehat{\mathbb{C}}$. A circle domain has complementary components that are closed round disks or singletons. The unrestricted target is

$$\begin{gathered}\forall G \subset \widehat{\mathbb{C}} \text{ a domain}, \quad \exists f : G \longrightarrow D \text{ a conformal bijection}, \\ \forall K \in \operatorname{Comp}(\widehat{\mathbb{C}} \setminus D), \quad K \text{ is a disk or a point.}\end{gathered} \tag{1}$$

No connectivity bound, boundary smoothness, positive separation, uniform quasiconformal extension, or uniqueness is required. The quantifier over all complementary components includes uncountably many point components. The whole sphere is trivial. In a proper domain a Möbius change may place infinity in G; the source then preserves an exterior neighborhood and uses $f(z) = z + O(1/z)$ to exclude a constant limit.

There are three distinct questions: existence in (1), rigidity of conformal maps between circle domains, and construction with certified numerical error. Success at one does not imply the others. In particular, an approximation with small Cauchy–Riemann residual is not automatically injective, onto the intended image, or correct at every complementary end.

1.2 Pinned source and reading boundary

M149 denotes the 149th manuscript in the pinned `CONTENTS.md`, not a new numbering of mathematical conjectures. The public source is OpenAI's *Koebe's Circle-Domain Conjecture*, September 23, 2026, at repository commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a` [1]. Its source directory is

`preprints/Koebes-Circle-Domain-Conjecture-September-23-2026`.

The study directly examines the formulation, coordinate/compatibility definitions, finite-transfer statement and metric construction, normalized limits, end identification, Stokes estimate and proof, bad-pair exclusion, and final retained-test assembly. The broader path construction, probability fusion, and selection mechanism are tracked as source dependencies. This is not a line-by-line certification of every auxiliary topological or measurable selection argument, and no Lean proof was rebuilt in this study.

The L5 classification is an *applicability* score: the corpus's TR-item-343 addresses the same arbitrary-domain target and its analytic and all-component obstructions. It is not a solved-status score. The original corpus report explicitly leaves the unrestricted problem unresolved. OpenAI's repository likewise distinguishes verification stages across its collection. A released claim and an independently established theorem are not identified merely by changing representation.

1.3 Current literature and the new release

Classical finite-connectivity uniformization and the He–Schramm countable case are the background used by the source. Recent primary work establishes additional geometric subclasses. Karafyllia and Ntalampekos's September 25, 2026 preprint treats John domains and provides a transboundary-modulus criterion [2]. Their earlier work treats Gromov-hyperbolic domains [3]. These are different statements from unrestricted existence. Ntalampekos and Younsi's September 30 survey concerns rigidity and removability [4].

One logical dependency is especially important. Theorem 1.1 of Ntalampekos–Rajala's *Exhaustions of Circle Domains* assumes that the given domain already satisfies Koebe's conjecture and then supplies a suitable exhaustion [5]. It is not an independent proof of (1). Arbitrary interior exhaustions can

fail. The OpenAI source uses the exhaustion result as a consequence after its own existence argument, which is the appropriate logical order. This report does not replace that existence argument by the exhaustion theorem.

2 Source Architecture and Ram Interfaces

2.1 The source's proof chain

The source collapses each complementary component to a label in a compact totally disconnected set B in a quotient sphere X. A coordinate is either a bounded real function with finite Dirichlet energy or a locally real function with an additive period one around a specified end. Its differential is single-valued. Nonsingular ends have an oscillation budget $a_i \in \ell^2(B)$; the sum over arbitrary B means the supremum of finite subsums. Consequently its positive support is countable.

The source's condition (P), Definition `def:compatibility`, is a level-set nonintersection assertion involving each new coordinate and finitely many earlier coordinates, including repeated labels, outside a countable flag set. It is a geometric condition, not an algebraic rank test. It must not be erased when coordinates are compressed or recombined.

Source step	Ram-compatible operation	Untransferred requirement
Quotient and compatible levels, Section 1	Represent local fields and retain end/period labels	Moore decomposition, generic-level topology, and (P).
Finite transfer, Section 2	Exact moments, wedge identities, physical energy and metric optimization	Simultaneous extension with exact core values and vanishing tail/oscillation errors.
Barriers, Section 3	Positive energy certificates and explicit norm budgets	Admissible path coding, incidence, multiplicity, and splicing.
Positive laws, Section 4	Finite primal/dual moment tests and bounded barycenters	One law valid on all dust; pointwise admissibility on reference-null sets.
Selection, Section 5	Maintain countable provenance and fixed witnesses	Countability and persistence through all compatible extensions.
Streams and Stokes, Section 6	Closed-form periods, sharper boundary variation, full physical error budgets	Construction of arbitrarily small compatible streams and positive protrusion lower bounds.
Final limit	Full-component compactness and inverse containment tests	All components, all scales, and a nonconstant conformal limit.

2.2 Actual corpus inputs

TR250 [9], equations `eq:moments` and its compatible-form constructions, starts from a finite physical space and transfers its differential complex and mass matrix to moment coordinates. It explicitly does not infer a continuum theorem from finite invertibility. The revised Foundation version of the same material [10] makes the same boundary explicit. TR-item-343 [8] provides full-boundary Cauchy estimates, period repair, inverse compact containment, and full-cluster compactness. These are the most direct sources for this manuscript.

The relevant classical foundations include compatible finite element complexes [7] and modulus–probability duality [6]. Ram coordinates are a way to organize these structures with explicit physical

metrics, not a new name that changes their hypotheses or establishes historical priority. The other requested corpus directories were inventoried and routed through evidence maps; no assertion is made that every theorem in every book or application report was read or validated.

3 Physical Moment Representation and Exact Periods

3.1 Coordinates, closure, and energy

Let V_h be a specified finite-dimensional space of real one-forms on a fixed smooth finitely connected region, with basis $\psi_1, \dots, \psi_N$. For $\alpha_h = \sum c_j \psi_j$, define moments

$$m_i = \ell_i(\alpha_h), \quad m = Ec, \quad E_{ij} = \ell_i(\psi_j), \quad c = Rm, \quad R = E^{-1}. \tag{2}$$

Square invertibility is assumed. A redundant bank needs a specified left inverse on its range and a range-consistency test. If $H_{ij} = \int \langle \psi_i, \psi_j \rangle_g \, \mathrm{d}A_g$, then

$$\|\alpha_h\|^2_{L^2(g)} = c^T H c = m^T R^T H R m. \tag{3}$$

Raw Euclidean coefficient norm need not equal physical energy.

For a compatible complex with differential matrices D^k, invertible coordinate maps E_k give $D^k_R = E_{k+1} D^k E_k^{-1}$, and $D^{k+1}_R D^k_R = 0$ follows from the original complex. This preserves an existing identity; it does not create missing topology. For the stream application, select the closed-form subspace before imposing periods. Local polynomial primitives alone do not test circulation around a hole.

3.2 Minimum-energy period repair

Let $\gamma_1, \dots, \gamma_q$ form a complete homology cycle bank for the fixed finite region, and let $P_{ij} = \int_{\gamma_i} \psi_j$ on a basis of closed forms. Suppose $H > 0$ and P has full row rank. A target period vector may have one component equal to one and the others zero, as in the source.

Theorem 3.1 (Optimal finite physical correction)**.** *Given coefficients c and target p, write $e = p - Pc$ and $S = PH^{-1}P^T$. Then $S > 0$, and the unique correction minimizing $\delta^T H \delta$ subject to $P\delta = e$ is*

$$\delta = H^{-1} P^T S^{-1} e, \qquad \|\textstyle\sum \delta_j \psi_j\|^2_{L^2} = e^T S^{-1} e. \tag{4}$$

The corrected closed form has exactly the prescribed periods.

Proof. For nonzero v, $P^T v \neq 0$, so $v^T S v > 0$. Formula (4) satisfies $P\delta = e$. Every other feasible correction is $\delta + z$ with $Pz = 0$. Since $z^T H \delta = z^T P^T S^{-1} e = 0$, its energy is $\delta^T H \delta + z^T H z$, strictly larger unless $z = 0$. Substitution gives the displayed energy. Closedness is preserved by the linear combination; completeness of the cycle bank gives the asserted global period condition. □

This refines the corpus's finite period repair by making the correction optimal in a declared physical norm. It is classical constrained quadratic minimization specialized to the Ram representation, not a new universal inverse. In moment coordinates replace H by $R^T H R$ and P by PR; the resulting physical form is identical. If the computed matrix or period integrals are uncertain, their uncertainty must be enclosed before claiming exact satisfaction. Full row rank cannot be inferred from the number of rows.

For an exact finite example, take

$$H = \operatorname{diag}(2, 3, 5), \quad P = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}, \quad e = \begin{pmatrix} 1/4 \\ -1/5 \end{pmatrix}.$$

Then $\delta = (3/10, -1/20, -3/20)^T$, $P\delta = e$, and $\delta^T H \delta = 3/10$. The kernel vector $(-1, 1, -1)^T$ is H-orthogonal to δ, checking the minimization identity directly.

Remark 3.2 (Compatibility is a separate obligation). The source's (P) depends on level sets. An energy-optimal correction or arbitrary linear recombination need not preserve (P), masks, exact agreement on a retained core, or boundary bands. Theorem 3.1 is an implementation and audit tool on a fixed finite model. It is not a license to substitute repaired coordinates into the global selection without checking those additional conditions. Likewise, a finite cycle bank is not automatically complete for an infinitely connected domain.

4 A Sharper Stream–Stokes Estimate

4.1 Geometry and hypotheses

Let C be a positively oriented smooth Jordan curve whose bounded interior lies in $B(0, L)$. Remove finitely many pairwise disjoint closed round disks $D_j = \overline{B}(c_j, r_j)$ compactly inside C, and write Ω for the remaining region. First take positive radii, so all boundaries are smooth. Let D_b be distinguished. Let α be a smooth closed real one-form on a neighborhood of $\overline{\Omega}$, with period one around D_b and period zero around every other hole, using counterclockwise hole orientation. Locally $\alpha = \mathrm{d}u$. Near a nonsingular hole its zero period gives a single-valued branch u_j, with $A_j = \operatorname{osc}_{\partial D_j} u_j$. Set

$$\phi(w) = \operatorname{dist}(w, D_b), \quad E = \|\alpha\|_{L^2(\Omega)}, \quad A = \operatorname{area}(\Omega), \quad a = \Big(\sum_{j \neq b} A_j^2\Big)^{1/2}.$$

This is the finite geometry in the source's Lemma `lem:stream-stokes`. If only cluster oscillations are available, apply the argument on concentric offsets and pass to their limits under the source's collar control. That passage is stated explicitly below.

Lemma 4.1 (Exact circular variation). *For $j \neq b$, $\operatorname{Var}_{\partial D_j} \phi = 4r_j$ and*

$$\left| \int_{\partial D_j} \phi \, \mathrm{d}u_j \right| \leq 2 r_j A_j. \tag{5}$$

Proof. Let $d = |c_j - c_b| \geq r_j + r_b$. On the other circle, $\phi = |w - c_b| - r_b$. After rotating the parameter its distance term is $\sqrt{d^2 + r_j^2 + 2 d r_j \cos t}$. It decreases from $d + r_j$ to $d - r_j$ on $[0, \pi]$ and increases on $[\pi, 2\pi]$. Its total variation is $2[(d + r_j) - (d - r_j)] = 4r_j$. Subtract the midpoint of the range of u_j. Its absolute value is then at most $A_j/2$. Integration by parts on the closed circle gives $\int \phi \, \mathrm{d}u_j = -\int (u_j - c) \, \mathrm{d}\phi$ and proves (5). The same variation formula follows by continuity in the tangential limiting case. The coefficient $2r_j$ is sharp if only the oscillation of a free single-valued boundary function is prescribed: smooth approximations to opposite constant values on the two monotonicity arcs approach the bound. This sharpness assertion does not impose a simultaneous fixed energy budget. □

Theorem 4.2 (Refined finite stream bound). *Under these hypotheses,*

$$\left| \int_C \phi \alpha \right| \leq \sqrt{A}\, E + 2 \sum_{j \neq b} r_j A_j, \tag{6}$$

$$\leq \sqrt{A}\, E + 2 \sqrt{L^2 - r_b^2 - A/\pi}\, a, \tag{7}$$

$$\leq \sqrt{L^2 - r_b^2}\, \sqrt{\pi E^2 + 4a^2}. \tag{8}$$

In particular, the simpler bound $\sqrt{\pi} LE + 2La$ is valid.

Proof. Stokes' theorem, with the inner-boundary signs reversed relative to C, gives

$$\int_C \phi\alpha = \int_\Omega \mathrm{d}\phi \wedge \alpha + \sum_j \int_{\partial D_j} \phi\alpha.$$

Closedness removes a $\phi\,\mathrm{d}\alpha$ term. The distinguished boundary has $\phi = 0$. Since $|\nabla\phi| = 1$ almost everywhere outside D_b, the bulk term is at most $\sqrt{A}E$. Lemma 4.1 treats the other holes, proving (6). Packing gives $A + \pi\sum_j r_j^2 = \mathrm{area}(\mathrm{int}\, C) \le \pi L^2$. Cauchy–Schwarz on the nonsingular holes proves (7). Finally put $t = A/\pi$ and $M = L^2 - r_b^2$. Then

$$\sqrt{\pi t}\, E + 2\sqrt{M - t}\, a \le \sqrt{M}\sqrt{\pi E^2 + 4a^2}, \quad 0 \le t \le M,$$

which is a second two-dimensional Cauchy–Schwarz inequality. □

The original source uses $|\,\mathrm{d}\phi| \le |\,\mathrm{d}w|$ and a full oscillation bound, leading to $2\pi r_j A_j$. The midpoint and exact variation argument reduces this coefficient by a factor of π. This is a definite improvement of that displayed estimate. The bulk area factor also uses actual occupied area rather than replacing it immediately by πL^2. No claim is made that these constants are historically new or that the coupled estimate is sharp under all joint constraints on a genuine stream.

4.2 Offset circles and the limiting estimate

In the source, boundary values may be handled by shrinking collars rather than by extending the uniformizing map to the entire boundary. For each finite completion choose disjoint slightly enlarged circles, all contained in the domain. On the distinguished offset, $\phi = \rho$ and its integral against the period-one form equals ρ. On another offset the exact variation is $4(r_j + \rho_j)$ and its oscillation is at most $A_j + o(1)$. Let offsets tend to zero. Smoothness on compact subregions and the finite energy/collar hypotheses give (6). A radius-zero hole can be treated through this limiting procedure when its hypotheses hold; no finite-energy nonzero-period form on an isolated full punctured disk is silently assumed.

For the source's diagonal sequence, let the transferred energies obey $E_n \le E_0 + o(1)$, the positive-excess oscillation estimate imply $a_n \le a_0 + o(1)$, and $r_{n,b} \to r_b$. The common target radius L then yields the conditional refinement

$$\limsup_n \left| \int_{C_n} \mathrm{dist}(w, D_{n,b})\, \mathrm{d}u_n \right| \le \sqrt{L^2 - r_b^2}\sqrt{\pi E_0^2 + 4a_0^2}. \tag{9}$$

Using $L\sqrt{\pi E_0^2 + 4a_0^2}$ avoids needing a positive limiting radius. This conclusion assumes the source's transfer and period data; it does not construct them.

4.3 Quantitative protrusion exclusion

In the source's bad-pair argument the unsmoothed boundary integral has a positive lower bound d_0. Smoothing and convergence reduce it to a strictly positive bound such as $d_0/2$. If a rigorous lower enclosure is $\underline{I}$ and a rigorous upper enclosure from (8) is $\overline{B}$, the contradiction requires the single strict inequality

$$\underline{I} > \overline{B}. \tag{10}$$

This can tolerate a larger oscillation budget than the original $\sqrt{\pi}LE_0 + 2\pi La_0$ criterion. It does not weaken the essential requirement that compatible small streams exist for every bad pair.

There is an explicit robustness estimate for the lower enclosure. On a fixed source contour let (f, D, α) be replaced by $(\widehat{f}, \widehat{D}, \widehat{\alpha})$ with $\sup_C |f - \widehat{f}| \le \eta_f$, $d_H(D, \widehat{D}) \le \eta_D$, and $\|\alpha - \widehat{\alpha}\|_{L^1(C)} \le \eta_\alpha$. Then

$$|I - \widehat{I}| \le (\eta_f + \eta_D) \|\alpha\|_{L^1(C)} + \sup_C \operatorname{dist}(\widehat{f}, \widehat{D}) \eta_\alpha. \tag{11}$$

Indeed distance functions change by at most the Hausdorff error, and are 1-Lipschitz in their argument; subtract the two integrals and use the triangle inequality. Period errors, topology errors, and unobserved trace tails are not absorbed into a floating-point evaluation of η_α.

5 Metric Optimization with a Distortion Budget

5.1 Rederivation of the source's local mechanism

In a reference orthonormal chart write $v_i = \nabla u_i$ and $S = \sum_i v_i v_i^T \succeq 0$. The source's Step 5 in finite transfer chooses $g_\tau = S + \tau I$. Its summed energy density is

$$\operatorname{tr}(g_\tau^{-1} S) \sqrt{\det g_\tau} = \frac{2\lambda_1\lambda_2 + \tau(\lambda_1 + \lambda_2)}{\sqrt{(\lambda_1 + \tau)(\lambda_2 + \tau)}} \longrightarrow 2\sqrt{\det S}. \tag{12}$$

Here $\lambda_1 \ge \lambda_2 \ge 0$. The Cauchy–Binet identity gives

$$\det S = \sum_{i<j} \det(v_i, v_j)^2.$$

Thus small wedge products allow small total energy in a newly selected metric. The source supplies a dominating integrable function and subsequently glues the metric to the retained-core metric. Both steps are needed; a pointwise limit by itself is insufficient.

In two dimensions conformal rescaling of a metric does not change Dirichlet energy. We may normalize its matrix G to determinant one, so the local objective becomes $\operatorname{tr}(G^{-1}S)$. Without a distortion cap its infimum is $2\sqrt{\det S}$; in rank one the zero infimum requires unbounded anisotropy and is not attained by a positive definite metric.

Theorem 5.1 (Exact distortion-constrained energy)**.** *For $K \ge 1$ define*

$$F_K(S) = \min\{\operatorname{tr}(G^{-1}S) : G > 0,\ \det G = 1,\ \kappa(G) \le K\}.$$

Then $F_K(0) = 0$. Otherwise

$$F_K(S) = \begin{cases} 2\sqrt{\lambda_1\lambda_2}, & \lambda_2 > 0,\ \lambda_1/\lambda_2 \le K, \\ \lambda_1/\sqrt{K} + \lambda_2\sqrt{K}, & \lambda_2 = 0 \text{ or } \lambda_1/\lambda_2 > K. \end{cases} \tag{13}$$

For positive semidefinite S, T,

$$|F_K(S) - F_K(T)| \le (\sqrt{K} + 1/\sqrt{K}) \|S - T\|_{\mathrm{op}}. \tag{14}$$

Proof. The eigenvalues of G are a, a^{-1} with $1 \le a \le \sqrt{K}$. For fixed a, align the eigenvector of a with that of λ_1. This is optimal: rotating by angle θ adds $(\lambda_1 - \lambda_2)(a - a^{-1}) \sin^2\theta \ge 0$ to the aligned trace. It remains to minimize $\lambda_1/a + \lambda_2 a$ on that interval. Its derivative is $-\lambda_1/a^2 + \lambda_2$, giving the stationary value $a = \sqrt{\lambda_1/\lambda_2}$ when allowed, and the capped endpoint otherwise. This proves (13) including rank one. For any feasible G, $|\operatorname{tr}(G^{-1}(S-T))| \le \operatorname{tr}(G^{-1}) \|S - T\|_{\mathrm{op}}$, and $\operatorname{tr}(G^{-1}) = a + a^{-1} \le \sqrt{K} + 1/\sqrt{K}$. Apply this inequality at a minimizer for S and then at a minimizer for T to obtain both directions. □

For $S = \operatorname{diag}(9, 1)$, a cap $K = 4$ gives $F_K = 13/2$, whereas $K = 9$ gives 6. For $S = \operatorname{diag}(4, 0)$ and $K = 4$, the minimum is 2, although the unrestricted infimum is zero. This is an exact obstruction to demanding arbitrary small energy while imposing a fixed finite metric condition number. It is not an obstruction to the source construction, which does not impose such a uniform cap.

5.2 Panel certification and the smoothness boundary

If a Ram panel reconstruction gives $\widehat{S}$ with $\left\|S-\widehat{S}\right\|_{\mathrm{op}} \le \delta$ pointwise, (14) encloses the local optimum with error at most $(\sqrt{K}+1/\sqrt{K})\delta$. Integrating a proved panel envelope gives a bound for the integral of pointwise minima. Gram matrices and derivative couplings must be computed in the reference physical chart, with the correct Jacobian under curved pullback. Certified quadrature and unrepresented derivative tails contribute separate errors.

This local theorem does not assert that an arbitrary measurable optimizer can be used as the source's smooth global metric. Eigenvector changes, mask boundaries, interpolation to the fixed core, and a possible loss in the cap under smoothing require additional control. A practical variant can allow a small cap enlargement and allocate explicit collar energy. Proving that the necessary collar and approximation bounds are uniform through the source's full selection remains a separate research task.

6 The Infinite-Component Limit Must Remain Visible

6.1 A robust Ram-compatible sufficient criterion

The following corpus criterion [8, `thm:robust`] is included with proof because it explains precisely why finite circle fitting alone cannot replace the source's end-selection mechanism. For compact sets define $d^+(D,E)=\sup_{z\in D}\operatorname{dist}(z,E)$ and let $N_\delta(E)$ be the open δ-neighborhood.

Theorem 6.1 (Robust full-component compactness)**.** *Suppose nonempty compact sets $E_j \subset \overline{B}(0,R)$ converge in Hausdorff distance to E, and $\eta_j \to 0$. Assume*

$$\forall \epsilon > 0\ \exists \delta > 0, J\ \forall j \ge J \tag{15}$$

such that every connected component U of $N_\delta(E_j)$ with $\operatorname{diam} U \ge \epsilon$ has a closed disk or point D satisfying $U \subset N_\epsilon(D)$ and $d^+(D,E_j) \le \eta_j$. Then every component of E is a disk or a point.

Proof. Fix a component C of E of diameter $d>0$ and choose $\epsilon_m \downarrow 0$ with $\epsilon_m < d$. Choose increasing $j_m \ge J_m$ so $d_H(E_{j_m},E) < \min(\delta_m, 1/m)$. The connected set C lies in one component U_m of $N_{\delta_m}(E_{j_m})$, and $\operatorname{diam} U_m \ge d$. Obtain templates D_m. Their directed error bounds keep their centers and radii bounded. A subsequence converges to a disk/point D. Domination gives $C \subset D$; directed error and Hausdorff convergence give $D \subset E$. The connected set D meets C, so maximality of the component gives $D \subset C$. Thus $D = C$. Components of diameter zero are singletons. □

The criterion permits geometry error in E_j and does not require those finite approximations already to have circular components. It is sufficient, not asserted necessary. In particular, it does not supply a construction meeting (15) for every original domain.

6.2 Two exact failure tests

Two unit disks centered at $\pm(1+1/j)$ have disjoint components and converge to two tangent disks. Their union in the limit is connected but is not a disk. Separately, the point sets $\{k/j : 0 \le k \le j\}$ converge to the segment $[0,1]$ with Hausdorff distance $1/(2j)$. Every finite component is a singleton, yet the limiting component is not. These examples are not counterexamples to Koebe uniformization of a fixed domain. They refute the shortcut from round finite components to round limiting components.

The order in (15) is essential. Choosing a different $\delta_j \downarrow 0$ at every index can isolate every point in the second example and still miss the limiting segment. A finite set of accepted scales does not prove all scales. Likewise, controlling a finite collection of large holes leaves the combined effect of omitted small holes and point dust open.

6.3 How the two approaches can be assembled

The source's route obtains a countable marked set, proves collapse outside it, and excludes every marked protrusion by a persistent test. The Ram full-cluster route instead assumes a uniform domination certificate for the whole approximating complement. These are alternative sufficient mechanisms, not interchangeable claims. A conditional computational study can combine:

1. a nonconstant locally uniform limit of genuinely conformal injections;
2. a full-complement Hausdorff enclosure and inverse compact containment identifying the actual image rather than a guessed target;
3. either the source's proved all-end exclusion hypotheses, or a verified all-scale condition of Theorem 6.1.

Under those premises the image is a circle domain. A moment transformation does not establish any missing premise. The source's normalization is a particularly effective noncollapse condition and should be retained rather than replaced by a numerical near-normalization without an error bound.

7 A Reproducible Finite Workflow

For a fixed finite completion the following workflow has precise acceptance conditions; it is not an algorithm claimed to terminate for every domain.

1. Record the actual region, all holes, contour orientations, metric, closed-form space, and a complete finite period bank.
2. Construct moment maps and physical Gram matrices. Establish their rank and conditioning with interval or exact arithmetic, not just a small floating residual.
3. Reconstruct the selected fields with explicit full-tail bounds. Repair period errors by Theorem 3.1 when its premises hold; check masks and compatibility separately if the source construction uses the repaired field.
4. Enclose area, radii, nonsingular oscillations, and energy, then apply (6) or (8). If varying the metric, apply (13) with declared cap and include smoothing/gluing costs.
5. Enclose the boundary integral independently using all retained contour panels and (11). Accept a bad-pair exclusion only when (10) holds strictly.
6. Save the assumptions and certificate. A zero or negative margin is inconclusive, not evidence that a noncircular limit exists.

The local program `../_audit/check_M149.py` checks the finite period example exactly with rational arithmetic, verifies the displayed metric examples, and checks the circle-extrema arithmetic. Its JSON output records a pass. The analytic proofs above, not sampled numerical tests, establish the inequalities for all parameters. The script is not an implementation of the source's path-law construction or a verifier of (1). No speedup, mesh complexity, or experimental uniformization success is asserted without an executed comparison.

8 Advantages, Costs, and Research Directions

8.1 Advantages relative to the source

The sharper circle variation reduces a concrete boundary coefficient while preserving the source's periods and geometry. Joint area–radius accounting avoids maximizing correlated budgets separately. A distortion cap makes the price of the source's anisotropic energy reduction explicit. Physical moment norms and optimal period repair expose the finite rank and topology conditions needed for implementation. Full-tail and all-component tests help distinguish an accepted finite certificate from an unjustified limit.

These gains are in precision, transparency, and controllability of selected steps. They do not replace the hard arbitrary-domain topology or prove that the source's global selection is valid. The elementary ingredients used here are classical; the contribution is their explicit synthesis and the proved refinements at the identified source interfaces.

8.2 Costs and limitations

Moment banks can be ill-conditioned, derivative reconstruction needs regularity, and certified tail bounds may cost more than raw numerical evaluation. Exact periods are global constraints. Compatibility (P) and measurable path-law fusion are not finite Gram-matrix properties. A fixed distortion cap can prevent small energy even for rank-one fields. The full-cluster criterion may be too strong for a successful source sequence and cannot be inferred from finite samples. No verified new theorem of unrestricted uniformization is claimed in this report.

8.3 Concrete next questions

1. Integrate the improved Stokes constant into an explicit retained-test construction and quantify the reduction in depth needed for a chosen positive protrusion margin. The existence proof alone supplies no such complexity bound.

2. Establish a smooth metric approximation theorem with an explicit cap enlargement and collar-energy estimate for the source's evolving masks.

3. Design period-correcting operations that also preserve (P), exact core values, and boundary bands. The minimum-energy correction alone does not.

4. For John or other controlled domains, connect a published geometric criterion to computable all-scale Ram bounds before pursuing arbitrary connectivity. This offers a testable intermediate result without claiming that finite checks solve the universal problem.

5. Independently audit the source's positive-law alternative and fusion with all null-set, uniform-constant, and quantifier conditions retained. No finite discretization can substitute for a missing measure-theoretic step.

9 Conclusion

M149 is directly relevant to the current Ram corpus. A useful and rigorously bounded outcome is available: the stream boundary coefficient improves from 2π to 2, a coupled energy/oscillation budget follows, and the exact distortion-constrained metric optimum and minimum-energy period correction make important analytic costs explicit. Full-component compactness supplies a complementary sufficient limit criterion. These results improve selected steps and provide a controlled route for

further investigation while leaving the source's unrestricted existence claim and its deepest selection dependencies clearly separate.

A Source Locators and Evidence Boundary

The principal source locators are `def:coordinates`, `def:compatibility`, `thm:transfer`, `eq:transfer-metric-density`, `lem:normalized-limits`, `lem:end-components`, `lem:modulus-duality`, `lem:law-fusion`, `lem:stream-stokes`, `lem:bad-pair`, and `thm:retained-tests`. The local refinements use the actual finite geometry of the Stokes lemma and the actual metric objective of finite-transfer Step 5, not a title-level analogy.

The confidential source paths most directly used are:

- `C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/res-math/TR-item-343-astra.tex`, especially `eq:physical`, finite period repair, `thm:image`, `thm:robust`, and the degeneration examples.
- `C:/Users/mural/OneDrive/Documents/MyDocuments2026/res-rt/TechReps/FinalTechReps/TRs/TR250-Compatible-Variational-Ram-Panels/TR250-Compatible-Variational-Ram-Panels.tex`, physical spaces, `eq:moments`, compatible complexes, and cohomology.
- `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`, the finite moment representation and physical norm limitations.

The external literature checks are recorded in `../_audit/M149-public-literature.txt`. Source snapshot and artifact hashes are retained in the parent project. Private corpus passages were not sent as public search queries or uploaded to outside repositories.

References

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[9] M. Subbarao. *Compatible Variational Ram Panels*. TR250, September 2026. Confidential local corpus.

[10] M. Subbarao. Foundation revision of TR250, `TR-rev-1-old-tr250.tex`. Confidential local corpus; finite moment representation and physical norm passages.